

Arnav Sagar

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Education

- **Delhi Technological University(DTU), Delhi** New Delhi, India
B.Tech. – Software Engineering; CGPA: 8.75 Jul 2024 – Jun 2028
- **Kendriya Vidyalaya Sector-47, Chandigarh** Chandigarh, India
12th: 93% 10th: 94.4% 2022 – 2023

Work Experience

- **5G LAB, Department of Telecommunications, Delhi Technological University** New Delhi, India
Summer Research Intern (MEC, URLLC) [GitHub] Jun 2025
 - Engineered a **sub-25ms P95 end-to-end** on-device inference pipeline via per-channel entropy calibration, achieving **4× model compression** and **1.8× throughput** at <1.5% mAP loss; deployed per-person Lucas-Kanade optical flow on alternate frames, with IoU-based keypoint association across **CUDA, Metal MPS, and CPU** backends.
 - Designed a **self-supervised trajectory autoencoder** (~1,600 params, <0.1ms/track) augmenting 5 geometric IDS rules to surface **6 behavioural threat types** without labeled data; built a **simulated 5G URLLC transport layer** (1–5ms latency, 0.1% packet loss) for structured edge-to-core threat delivery. Presented at **PEC Chandigarh**.
- **AIMS-DTU** New Delhi, India
Research Intern (LLM Safety) [GitHub] Jun 2025 – Jul 2025
 - Architected a **3-stage LLM moderation pipeline** — **sub-10ms** regex-based filters, semantic embeddings, and a **fine-tuned DistilBERT** augmented with **XGBoost + LOF** on **TF-IDF** features to surface zero-day prompt injections outside the classifier’s training distribution.
 - Engineered a **risk-scored mitigation engine** across **5 harm categories** with **4-tier decision logic** using contextual **NER + POS proximity analysis**; validated against red-teaming prompts on **OpenChat-3.5** with **GPT-4o-mini** cross-validation, served via a production **FastAPI** server.

Relevant Projects

- **Thermal Super-Resolution Model** [GitHub] [Demo]
 - Engineered a **triple encoder cross-attention** fusion architecture for thermal super-resolution with two-stage optical-thermal alignment module and a **progressive Laplacian decoder** for multi scale reconstruction.
 - Integrated texture guided safety constraints and **physics aware losses** to suppress hallucinations, improving PSNR by **15.74%** and SSIM by **8.22%** compared to *SOTA model* as evaluated on dataset provided by **ISRO**.
- **Hybrid CTC-Attention Scene Text Recognition** [GitHub]
 - Designed a **CRNN** architecture that features a VGG-style CNN backbone, combining a 2-layer **BiLSTM** encoder with *Bahdanau attention* decoder for sequence modeling.
 - Implemented a joint CTC and Cross-Entropy loss with **beam search** to stabilize early training and refine character alignment, achieving **~89%** word accuracy on the **IIIT5K** benchmark after training on **9M** images.
- **Retrieval Augmentation System** [GitHub]
 - Built a multi-stage RAG pipeline over PDF corpora with semantic chunking, hierarchical small-to-big indexing, **HyDE-based query expansion**, and sub-question decomposition for complex queries.
 - Engineered hybrid dense-sparse retrieval (**FAISS + BM25**) with RRF fusion, cross-encoder reranking, **MMR** diversity selection, contextual compression, and **CRAG**-based hallucination suppression.
- **LLM Safety Shield** [GitHub] [API]
 - Deployed a *session-aware* jailbreak & toxicity classifier with **MiniLM** cosine similarity, **SHAP** token attribution, adversarial normalization (homoglyph, zero-width, base64), **Redis** caching and an **ONNX**-optimized inference.
- **Text Splitting and Embedding Visualizer** [GitHub] [Demo]
 - Built an interactive visualization tool for analyzing text chunking strategies used in **RAG pipelines** and embeddings with **Cosine semantic similarity** matrix, **recursive splitting**, metrics and overlaps.
- **AI vs Human Text Classification System** [GitHub]
 - Benchmarked *8+ classical and ensemble models* (LR, SVM, Naive Bayes, Decision Trees, RF + AdaBoost, kNN, ANN and attention based **LSTMs**) against a **fine-tuned RoBERTa**, achieving an accuracy of **0.9996**.

Technical Skills

- **Languages:** Python, C/C++, JavaScript, HTML/CSS, SQL
- **ML/AI Frameworks:** PyTorch, TensorFlow, ONNX, HuggingFace, OpenCV, Pandas, NumPy, LlamaIndex
- **Developer Tools & Backend:** Weights & Biases, Docker, Git, ROS2 (Humble), FastAPI, Streamlit

Achievements

- National Finalist, Smart India Hackathon 2025 Ahmedabad, India

Positions of Responsibility

- Coordinator, Business Bulls, DTU (Finance & Strategy Club)